

Yiqian Li

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Education

-
- University of Pennsylvania (UPenn), MSE in Robotics** Sept 2024 – May 2026
- **Coursework:** Machine Perception, Advanced Robotics, Learning in Robotics, Graph Neural Network
- The Chinese University of Hong Kong, Shenzhen (CUHKSZ), BS in Computer Science** Sept 2020 – June 2024
- **Coursework:** Advanced Machine Learning, 3D Data Processing, Computer Graphics, Parallel Programming, Operating System, Computer Architecture, Optimization II
- The University of California, Irvine (UCI)** Jan 2023 – June 2023
- **Coursework:** Design and Analysis of Algorithms, Introduction to Graphical Models, Project in AI, Introduction to Artificial Intelligence

Publications

-
- **Yiqian Li**, Wen Jiang, Kostas Daniilidis, *Next-Best-View Selection for Semantic and Dynamic 3D Gaussian Splatting*. Submitted to **International Conference on 3D Vision 2025**. [1]
 - Aojun Jiang, Miao Hao, **Yiqian Li**, Jiaqi Wang, Chunfeng Yue, Zongjie Huang, Ying Yang, Meng Ma, Qifeng, Lyu, Yu Sun, Zhuoran Zhang, *Automated Point-of-Care Semen Analysis Using Smartphone Imaging and Occlusion-Aware Multi-Object Tracking*. **IEEE Transactions on Automation Science and Engineering**. [2]
 - Haodong Jiang, Xiang Zheng, Yanglin Zhang, Qingcheng Zeng, **Yiqian Li**, Ziyang Hong, Junfeng Wu, *SCORE: Saturated Consensus Relocalization in Semantic Line Maps*. **IROS 2025**. [3]
 - Zhijing Li, Qiurai Fu, Zhijun Huang, Jianbo Yu, **Yiqian Li**, Yuanhao Lai, Yuchi Ma, Pinjia He, *Revisiting Log Parsing: The Present, The Future, and The Uncertainties*. **IEEE Transactions on Reliability**. [4]

Research Experience(Selected)

-
- Next-Best-View Selection for Semantic and Dynamic 3D Gaussian Splatting[1]**
- UPenn | Prof. Kostas Daniilidis  03/2025 – 05/2025
- Modeled next-best-view selection as Fisher Information maximization over SE(3) scene parameters, semantic features distilled from 2D semantic model, and deformation MLP weights.
 - Derived Jacobian-vector approximations of the Fisher matrix for 3D Gaussians and dynamic MLPs under gradient-based rendering.
 - Implemented a fully differentiable pipeline with custom CUDA kernels to support fast Fisher computation within real-time NBV selection loops.
- Uncertainty-Aware Monocular SLAM with Transformer-Based Depth Modeling**
- UPenn | Prof. Kostas Daniilidis  09/2024 – 12/2024
- Extended DROID-SLAM by integrating monocular depth prediction from pretrained models into the BA layer with uncertainty modeling.
 - Designed a Vision Transformer to estimate depth uncertainty, enabling uncertainty-weighted bundle adjustment.
 - Explored joint optimization of poses and depths under noisy monocular input.
- Semantic Line-Based Visual Relocalization[3]**
- CUHKSZ | Prof. Junfeng Wu  03/2024 – 08/2024
- Proposed a scene-agnostic visual relocalization framework using compact semantic 3D line maps extracted from posed RGB-D images.
 - Formulated a novel Saturated Consensus Maximization framework to solve robust Perspective-n-Line problems.
 - Designed a globally optimal solver with interval-based Branch-and-Bound acceleration.
- Sperm Quality Analysis under Smartphone Imaging[2]**
- CUHKSZ | Prof. Zhuoran Zhang  12/2022 – 09/2023
- Developed a pseudo-labeling pipeline using DBSCAN clustering on standard and unlabeled sperm images.
 - Applied Vision Transformer for semi-supervised classification to estimate morphologically normal sperm ratios.

- Optimized U-Net upsampling and improved occlusion-aware JPDA tracking to enhance sperm segmentation and motility analysis.

Design of a Log Detection and Analysis System for Enterprise Cloud Server[4]

CUHKSZ | Prof. Pinjia He 

09/2021 – 07/2022

- Built a semantic log parser with word2vec-based clustering and invariant mining to support root cause analysis.
- Evaluated and compared CNN, autoencoder, and LSTM models for log anomaly detection; deployed the system in a production cloud service.

Competition

DJI RoboMaster Robotics Competition

CUHK | 10/2020 – 05/2021

- Established a simulation environment in Gazebo to facilitate the intelligent robot development and testing.
- Evaluated QMIX and VDN multi-agent RL algorithms for decision-making across engineering, infantry, supply, and aerial robot units.
- Enabled route planning via A* algorithm, Dijkstra's algorithm, and a greedy heuristic.

Projects

Scalable Quadruped Imitation

Learning in Robotics | UPenn

- Reconstructed 3D animal motions using learned skeletons from monocular dog videos.
- Retargeted motions to Unitree Go2 via inverse kinematics and trained imitation policies in simulation.
- Designed custom RL rewards to improve gait realism and policy convergence.

Pick and Place Robot Competition

Introduction to Robotics | UPenn

- Designed an algorithm to align gripper poses with target objects exhibiting pose equivalence under varying object-centric coordinate systems.
- Implemented a collision-free path planning framework for constrained grasping tasks on a rotating platform.
- Developed a dynamic object motion prediction module to enable grasping of targets moving on a rotating disk.

Transient Rendering in Participating Media

Computer Graphics | CUHKSZ

- Integrated volumetric scattering into a differentiable transient renderer to model participating media.
- Generated transient images capturing photon trajectories for enhanced realism in extreme weather simulation.
- Applied principles of light transport and volume rendering for physically accurate visual effects.

Reinforcement Learning for Optimal Traffic Signal Control

Project in AI | UCI

- Designed reinforcement algorithms for traffic signal control to minimize vehicle wait times.
- Simulated traffic in SUMO with visualized vehicle flows.
- Implemented DQN, Double DQN, Dueling DQN, PPO with various exploration strategies and compared results.

Technologies

Languages: C/C++ | Python | CUDA | OpenMP | Java | MATLAB | SQL | ARM Assembly | AHDL | Verilog HDL

Libraries: PyTorch/LibTorch | TensorFlow | OpenCV | Open3D | PCL | Colmap | ROS | Diffusers | Eigen



香港中文大學 (深圳)
THE CHINESE UNIVERSITY OF HONG KONG, SHENZHEN
LONGGANG • SHENZHEN • CHINA • TEL (0086) 0755 8427 3626 • FAX (0086) 0755 8427 3691

學業成績表
TRANSCRIPT OF ACADEMIC RECORD

Name: LI, Yiqian
Student ID No.: 120090702

Chinese Name: 李亿芊
ID/Passport No.: 440106200210281827

Gender: Female
Date of Birth: 28 Oct 2002

Admitted in: September 2020
College: Harmonia College
School: School of Data Science
Major/Programme: Computer Science and Engineering
Mode of Study: Full-time
Degree/Diploma: Bachelor of Engineering

2020-21 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
CHI1000	Chinese	3.0	A-	29.0
CHM1001	General Chemistry	3.0	B	33.5
ENG1001	English Bridge Program (EBP)	3.0	A-	16.6
MAT1001	Calculus I	3.0	A-	32.5
PED1001	Physical Education	1.0	B	38.5
PHY1001	Mechanics	3.0	A	38.3
		Units Passed = 16.0	Term GPA = 3.581	

2020-21 Term 2

Course Code	Course Title	Units	Grade	% of A- and above
CSC1001	Introduction to Computer Science: Programming Methodology	3.0	B	38.7
CSC1002	Computational Laboratory	1.0	B+	30.5
ENG1002	English for Academic Purposes I	3.0	B+	29.1
GFH1000	In Dialogue with Humanity	3.0	A-	36.5
ITE1000	Information Technology	1.0	B	37.1
MAT1002	Calculus II	3.0	B+	27.8
PED1002	Fitness and Health	1.0	A-	38.6
STA2001	Probability and Statistics I	3.0	B+	29.6
		Units Passed = 18.0	Term GPA = 3.322	

2020-21 Summer Session

Course Code	Course Title	Units	Grade	% of A- and above
BIO1001	General Biology	3.0	A-	31.9
		Units Passed = 3.0	Term GPA = 3.700	

2021-22 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
CSC3001	Discrete Mathematics	3.0	B+	28.4
EIE2050	Digital Logic and Systems	3.0	B+	33.3
ENG2001	English for Academic Purposes II	3.0	A-	28.8
GFN1000	In Dialogue with Nature	3.0	A-	36.0
MAT2040	Linear Algebra	3.0	A	37.6





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Gender: Female
Date of Birth: 28 Oct 2002

2021-22 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
PHY1002	Physics Laboratory	2.0	B+	45.7
		Units Passed = 17.0	Term GPA = 3.565	

2021-22 Term 2

Course Code	Course Title	Units	Grade	% of A- and above
CSC3002	Introduction to Computer Science: Programming Paradigms	3.0	A-	36.0
CSC3050	Computer Architecture	3.0	A-	35.6
ENG2002S	English for Science & Engineering Communication	3.0	A-	34.6
MAT3007	Optimization	3.0	A	39.4
		Units Passed = 12.0	Term GPA = 3.775	

Honour(s)/Award(s): Dean's List 2021-22

2021-22 Summer Session

Course Code	Course Title	Units	Grade	% of A- and above
CSC3100	Data Structures	3.0	A-	36.8
		Units Passed = 3.0	Term GPA = 3.700	

2022-23 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
CSC3150	Operating System	3.0	A-	34.6
CSC3170	Database System	3.0	A-	37.8
DDA3020	Machine Learning	3.0	A	34.1
EIE3080	Microprocessors and Computer Systems	3.0	A-	28.8
GEB2001	Exploration of the Quantum World	3.0	A-	40.0
		Units Passed = 15.0	Term GPA = 3.760	

2022-23 Term 2

Courses taken at University of California, Irvine, United States accepted for transfer as equivalent to 15.0 CUHK(SZ) units

Honour(s)/Award(s): Dean's List 2022-23

2023-24 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
CSC4005	Parallel Programming	3.0	A	37.8
CSC4008	Techniques for Data Mining	3.0	A-	38.5
DDA4210	Advanced Machine Learning	3.0	A-	39.2
GED2107	Metaphysics and Philosophy of Mind	3.0	A-	37.7
MAT3220	Optimization II	3.0	A	36.4





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Name: LI, Yiqian
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Chinese Name: 李亿芊
ID/Passport No.: 440106200210281827

Gender: Female
Date of Birth: 28 Oct 2002

2023-24 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
STA2002	Probability and Statistics II	3.0	A-	32.8
Units Passed = 18.0 Term GPA = 3.800				

2023-24 Term 2

Course Code	Course Title	Units	Grade	% of A- and above
CSC4001	Software Engineering	3.0	B+	38.2
CSC4140	Computer Graphics	3.0	B+	29.2
ECE4520	3D Data Processing	3.0	A	36.4
GEA2000	Modern Chinese History and Culture	3.0	A	36.0
GEC3104	Tradition and Modernity: Cities in Chinese History	3.0	B+	35.6
STA4001	Stochastic Processes	3.0	B+	40.7
Units Passed = 18.0 Term GPA = 3.533				

***** Summary *****

	Total Units Passed	Cumulative GPA	Major GPA
CUHK(SZ)	120.0	3.613/4.000	3.629/4.000
Unit Exemption	15.0	-	-

Remarks:

Attended University of California, Irvine, United States as visiting student in Term 2, 2022-23 in partial fulfillment of the requirements of this degree. A complete academic record of this degree programme should include the academic transcript from the visiting university to be provided separately by the student.

Major GPA is the grade point average across a list of Major courses defined by the School Board, as being most relevant to the student's Major.



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Director of Registry Services

***** End of Transcript *****



**OFFICIAL TRANSCRIPT OF RECORD
UNIVERSITY OF CALIFORNIA, IRVINE
DIVISION OF CONTINUING EDUCATION**

Addressee

Student

Yiqian Li

Yiqian Li
63 Wellesley
Irvine, CA 92612
DOB: Oct 28 ID: 001099880 SSN: xxx-xx-xxxx

Course Number	Course Title	Date Began	Date Ended	Qtr Units	Grade
Courses Offered for Undergraduate or Graduate Credit					
COMPSCI XIC 132	COMPUTER NETWORKS	1/9/2023	3/17/2023	4.00	A
LPS XIC 29	CRITICAL REASONING	1/9/2023	3/17/2023	4.00	A+
COMPSCI XIC 171	INTRO ARTIFCL INTEL	1/9/2023	3/17/2023	4.00	A
COMPSCI XIC 179	ALGRTMS GRAPH MDLS	4/3/2023	6/9/2023	4.00	A
COMPSCI XIC 161	DES&ANALYS OF ALGOR	4/3/2023	6/9/2023	4.00	A
COMPSCI XIC 162	FORMAL LANG & AUTM	4/3/2023	6/9/2023	4.00	A+
COMPSCI XIC 175	PROJECT IN AI	4/3/2023	6/9/2023	4.00	A+

GPA for Undergraduate/Graduate Courses: 4.00

GPA for all courses: 4.00

***** END OF TRANSCRIPT *****

Printed 10/23/2023

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Grading System

A+, A, A- -	Excellent	P -	Pass
B+, B, B- -	Good	NP -	Not Pass
C+, C, C- -	Average	S -	Satisfactory
D+, D, D- -	Barely Passing	U -	Unsatisfactory
F -	Failure	NR -	No Report
I -	Incomplete	W -	Withdrew
N -	No Credit	IP -	In Progress
CE -	Continuing Education	R -	Repeat
(one unit equals ten contact hours of participation)			



This academic transcript is printed on special security paper with a blue background, the Seal of the University of California, and the signature of the Continuing Education Registrar. This is an official sealed instrument; a raised seal is not required.

Beatrice Han

Beatrice Han, Registrar and Director of Student Services.

GUIDE TO THE TRANSCRIPT OF RECORD

This transcript comprises a record of all work taken by the student through University of California Irvine, Division of Continuing Education (UCI DCE), except for any taken prior to September 1967. Course work taken prior to that date is stored and maintained by UCLA Extension (P.O. Box 24901, Dept. K., Los Angeles, CA 90024-0901, (310) 825-3708).

Course Numbering:

Course numbers issued by UCI Division of Continuing Education are classified as follows:

1 - 99	Lower Division
100 - 199	Upper Division
200 - 299	Graduate Level (Concurrent Enrollment only); requires a Bachelor's degree and consent of instructor
300 - 399	Professional credit courses in Education
400 - 499	Professional credit courses in fields other than education
700 - 799	Special enrichment courses for participants of precollege age
800 - 890	Non-credit courses. Many offer Continuing Education Units (see below)

Course Number Prefixes:

"X"	Courses offered only through Continuing Education
"XIC"	Courses offered through Concurrent Enrollment on the Irvine Campus *
"XI"	Courses offered through Continuing Education that are equivalent to that of main campus

Our entire program, course by course, is approved by the chairperson of an appropriate academic department and/or the Dean of an appropriate school of this campus. This scrutiny is in addition to that given by the Director of Continuing Education.

X300-Series courses: Professional Credit Courses in education; require employment as a teacher, teaching credential or equivalent background, unless otherwise noted.

X400-Series courses: Professional Credit Courses in fields other than education. Designed to provide professional credit for purposes of career or salary advancement and are recognized in business, industry, government, school districts, the professions and the arts. May or may not lead to an award such as a certificate.

800-Series courses: Non-credit Courses; many offering Continuing Education Units (CEUs). The CEU is a means for measuring and recording non-credit postsecondary-level study, useful where employers, relicensure agencies and other authorities require a specified number of hours of instruction for career advancement, and where non-credit study is acceptable. One CEU is awarded for ten contact hours of participation; less than ten hours are shown as fractional CEU.

* XIC Courses: These are regular UCI courses. Since students enrolled in XIC courses are not matriculated, their records are kept and maintained by UCI DCE. Students planning to use such courses to satisfy any UCI (or UC) requirements should first consult with and secure prior approval from the academic dean or chair of their major. Whether these courses are applicable to a degree program at UCI is subject to approval of the student's academic department. Units earned through Concurrent Enrollment may appear as transfer units but grades are not used in calculating the UC GPA. Students should consult with their institution regarding the transferability of units.

Applicability to a Degree Program: UCI Division of Continuing Education is not defined as a degree granting institution. Students intending to transfer UCI DCE course credit towards fulfilling a degree at another college or university should verify acceptance of the units with that institution. We recommend to students that they obtain permission from their academic advisors before taking UCI DCE courses. Policies vary and each institution reserves the right to determine the appropriateness of a candidate's previous work.

All credit is offered in quarter system units; semester equivalents are calculated below:

<i>Quarter:</i>	1	1 1/2	2	3	4	5
<i>Semester:</i>	2/3	1	1 1/3	2	2 2/3	3 1/3

To Test for Authenticity: The face of this document has the University Seal in blue ink. When photocopied, this seal will not duplicate.

Record of: Yiqian Li
Penn ID: 14112633
Date of Birth: 28-OCT
Date Issued: 16-JUN-2025

U N O F F I C I A L

Page: 1

The University of Pennsylvania

Level: Professional

Primary Program

Program: School of Engineering and Applied Science
Master of Science in Engineering
Division : School of Engineering and Applied Science
Masters
Major : Robotics

SUBJ	NO.	COURSE TITLE	CU GRD	R
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INSTITUTION CREDIT:

Fall 2024

CIS	5800	Machine Perception	1.00	A+
ESE	5140	Graph Neural Networks	1.00	A
MEAM	5200	Introduction to Robotics	1.00	A-
Ehrs:	3.00	GPA-Hrs: 3.00	QPts: 11.70	GPA: 3.90

Spring 2025

ESE	6500	Learning in Robotics	1.00	A
MEAM	5430	Performance, Stability and Control of UAVs	1.00	C
MEAM	6200	Advanced Robotics	1.00	B+
Ehrs:	3.00	GPA-Hrs: 3.00	QPts: 9.30	GPA: 3.10

Fall 2025

CIS	6800	Adv Tpc in Mach Percept	1.00	IN PROGRESS
ESE	5460	Principles of Deep Learning	1.00	IN PROGRESS
In Progress Credits	2.00			

***** TRANSCRIPT TOTALS *****

	Earned Hrs	GPA Hrs	Points	GPA
TOTAL INSTITUTION	6.00	6.00	21.00	3.50

TOTAL TRANSFER	0.00			
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OVERALL	6.00	6.00	21.00	3.50
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***** END OF TRANSCRIPT *****